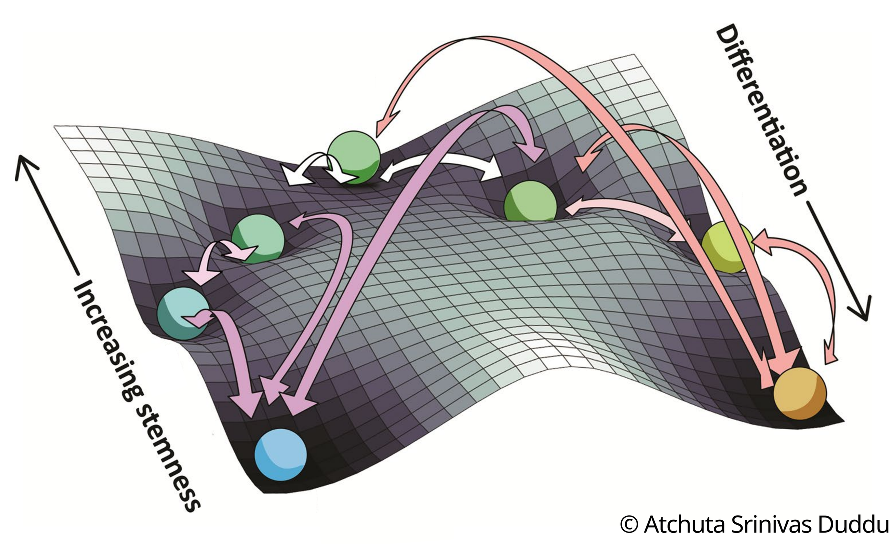
Harshavardhan BV[a,†], Sarah Adigwe[b,†], Tomas Gedeon[b,*], Mohit Kumar Jolly[c,*]
a IISc Mathematics Initiative, Indian Institute of Science, Bangalore, 560012, India
b Department of Mathematical Sciences, Montana State University, Bozeman, MT, 59717, USA
c Department of BioEngineering, Indian Institute of Science, Bangalore, 560012, India
† Equal contribution
E-mail: (HBV) bharshavardh@iisc.ac.in, (TG*) tgedeon@montana.edu, (MKJ*) mkjolly@iisc.ac.in,

Designing Network Topologies supporting Single-High States using Monotone Boolean Models

Funding:
PMRF
Prime Minister's Research Fellows

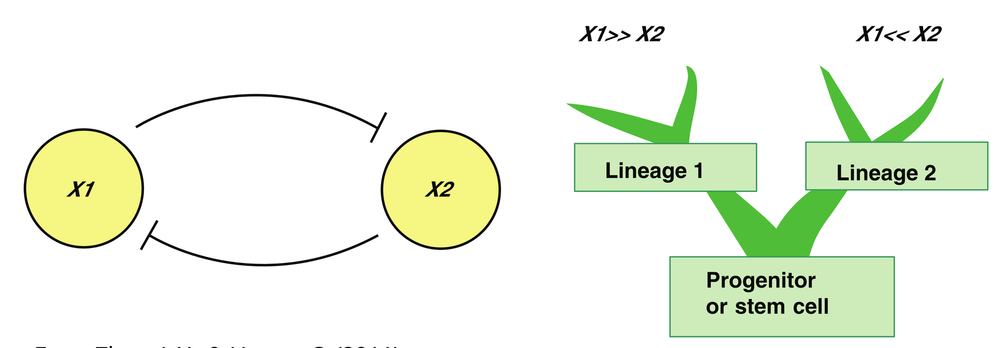
Introduction

Cell fate determination ≡ Waddington landscape
Valley: Terminally differentiated cell fate
Gene regulatory networks regulate trajectories



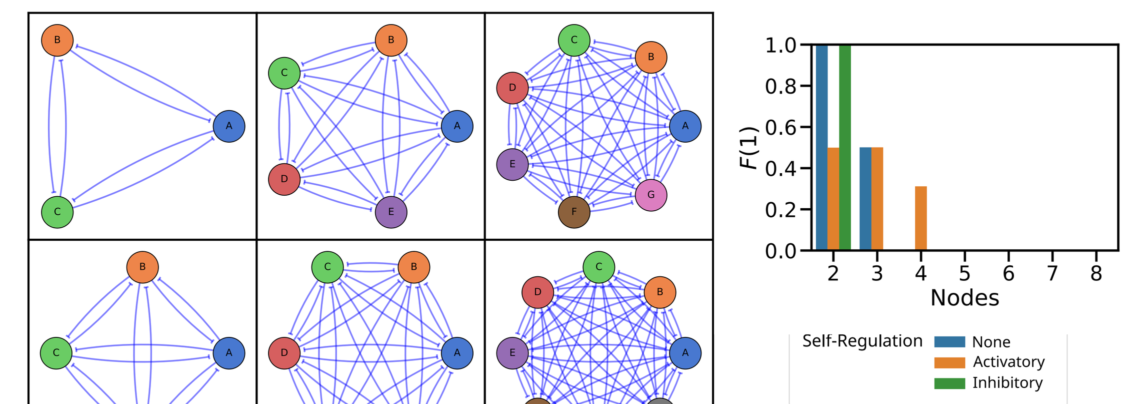
© Atchuta Srinivas Duddu

Cell-fate choices between two possible states
Toggle Switch: Mutual inhibition b/w
2 cell-state specific master regulator (transcription factors)



From Zhou, J. X., & Huang, S. (2011)

Terminal differentiation: Single-high states - only one master regulator expressed
Extending to n phenotypes - Toggle-n
Unable to capture fully differentiated states



From BV, H. et al. (2025)

WHICH NETWORKS CAN GIVE RISE TO FULLY DIFFERENTIATED STATES?

Methods

Definition

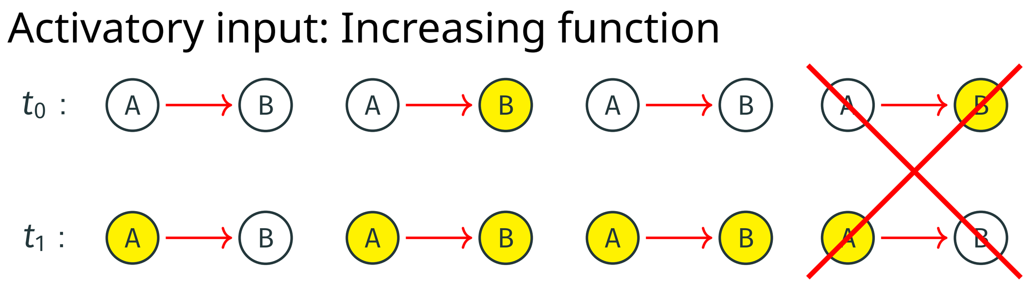
A Boolean function $f_i : B^k \to B$ is increasing (decreasing) w.r.t. input j if for any input $b = (b_1, \dots, b_k) \in B^k$

$f(b_1, \dots, b_{j-1}, 0, b_{j+1}, \dots, b_k) \le (\ge) f(b_1, \dots, b_{j-1}, 1, b_{j+1}, \dots, b_k)$.

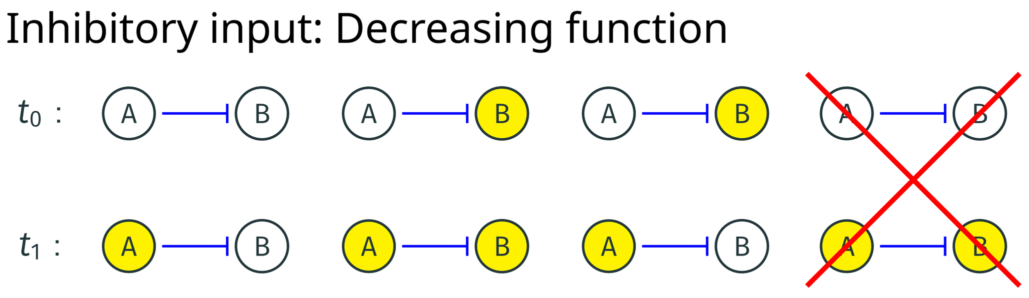
Definition

A Boolean function $f_i : B^k \to B$ is monotone if it is increasing or decreasing with respect to each input $j = 1, \dots, k$.
In this case, f_i is called a monotone Boolean function.

Activatory input: Increasing function



Inhibitory input: Decreasing function

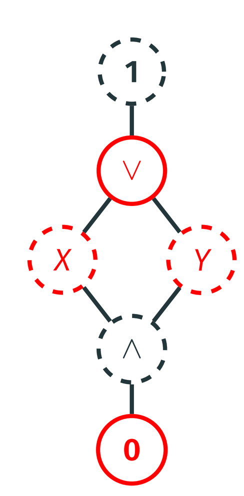


Inputs		MBFs					
X	Y	**0**	$\wedge$	**X**	**Y**	$\vee$	**1**
0	0	0	0	0	0	0	1
0	1	0	0	0	1	1	1
1	0	0	0	1	0	1	1
1	1	0	1	1	1	1	1

Definition

$U(b) := \{ f \in MBF(k) \mid f(b) = 1 \}$
$L(b) := \{ f \in MBF(k) \mid f(b) = 0 \}$

L(11)	$\{\mathbf{0}\}$	1
L(10)	$\{\mathbf{0}, \wedge, Y\}$	3
L(01)	$\{\mathbf{0}, \wedge, X\}$	3
L(00)	$\{\mathbf{0}, \wedge, X, Y, \vee\}$	5
U(11)	$\{\mathbf{1}, \vee, X, Y, \wedge\}$	5
U(10)	$\{\mathbf{1}, \vee, X\}$	3
U(01)	$\{\mathbf{1}, \vee, Y\}$	3
U(00)	$\{\mathbf{1}\}$	1



Definition

A Boolean model $f : B^n \to B^n$, $f = (f_1, \dots, f_n)$, is monotone if for every node v_i the function f_i is monotone.

$$MBM := MBF(k_1) \times MBF(k_2) \times \dots \times MBF(k_n)$$

Theorem

Consider network $N = (V, E, \delta)$ and Boolean state $b = (b_1, \dots, b_n) \in B^n$.
If b is a steady state, $f(b) = b$.
Then b is steady state of *monotone Boolean models* $f = (f_1, f_2, \dots, f_n)$ where

$$f_i \in I_i(b)$$

where

$$I_i(b) = \begin{cases} U(\,T_i(b_{S(i)})\,) & \text{if } b_i = 1 \\ L(\,T_i(b_{S(i)})\,) & \text{if } b_i = 0 \end{cases} \quad \text{and} \quad T_i^j(b) = \begin{cases} b_j & \text{if } \delta(v_j, v_i) = 1 \\ \neg b_j & \text{if } \delta(v_j, v_i) = -1 \end{cases}$$

A steady state is more prominent if more models support it.

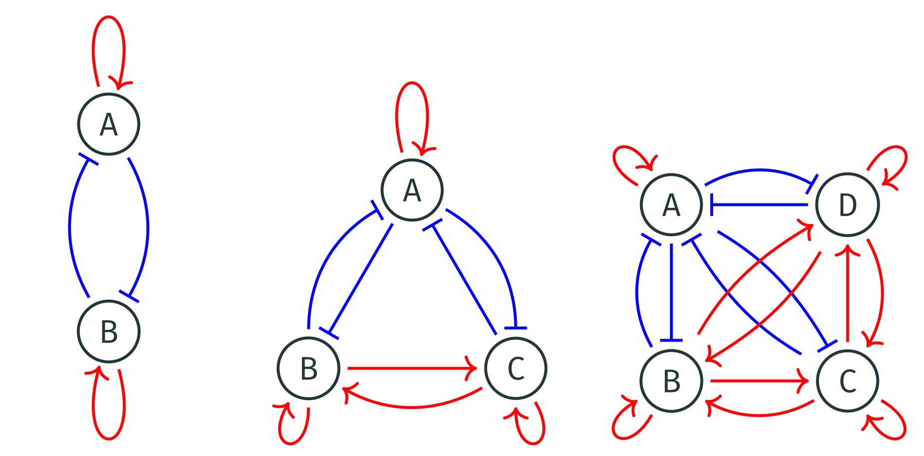
Theorem

Consider q Boolean states $b^1, \dots, b^q \in B^n$.
Then the multistability between these steady states is supported by *monotone Boolean models* $f = (f_1, f_2, \dots, f_n)$ where

$$f_i \in \bigcap_{j=1}^{q} I_i(b^j)$$

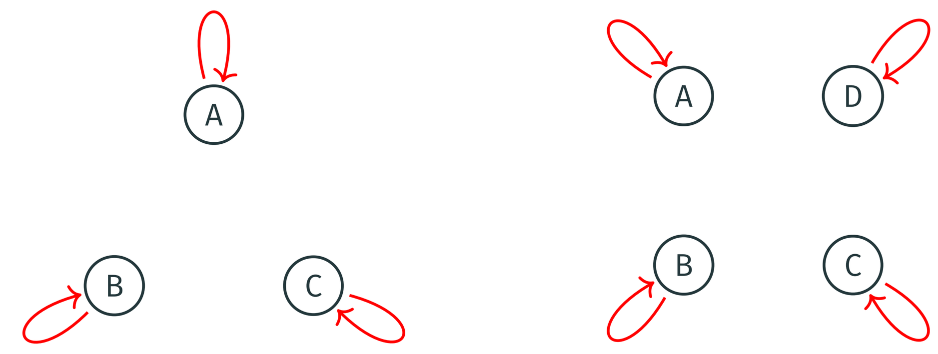
Networks supporting most Single-High

Maximum no. of *MBMs* for $e^1 := (1, 0, \dots, 0)$



2-Team networks: $G_1 = \{v_1\}$, $G_2 = \{v_2, \dots, v_n\}$

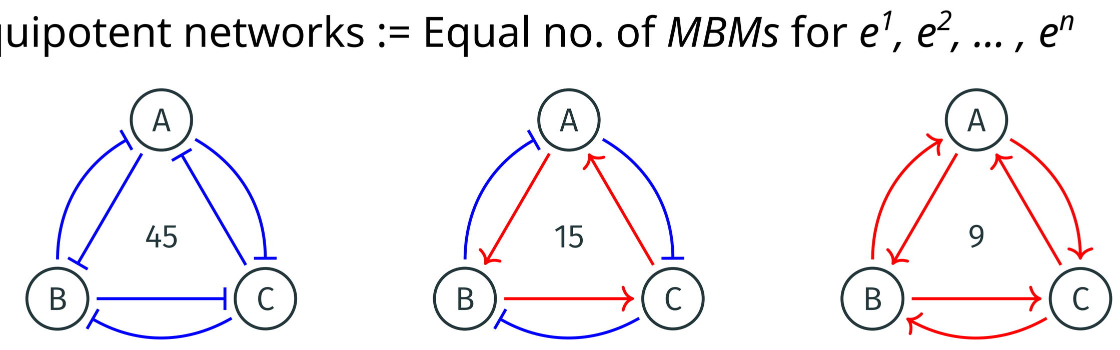
Maximum no. of *MBMs* for $e^1, e^2, \dots, e^n$ $(n>2)$



Disconnected networks with positive self-edge

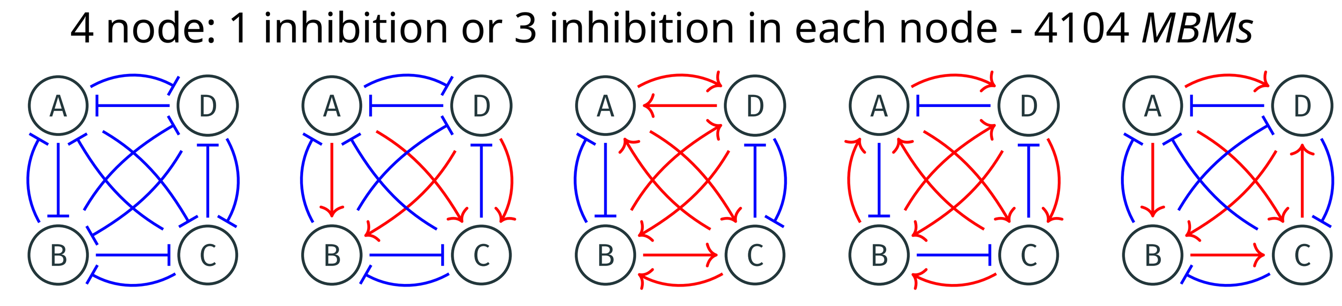
Networks supporting Equal Single-High

Equipotent networks := Equal no. of *MBMs* for $e^1, e^2, \dots, e^n$

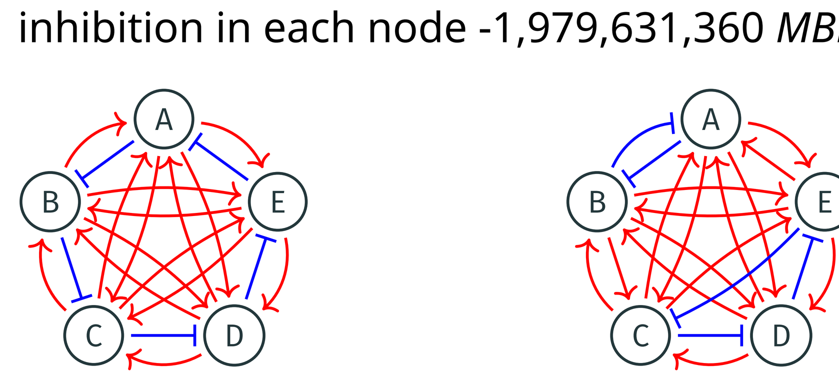


Inset: No. of MBMs supporting each e^i

Maximum among equipotent:
4 node: 1 inhibition or 3 inhibition in each node - 4104 *MBMs*

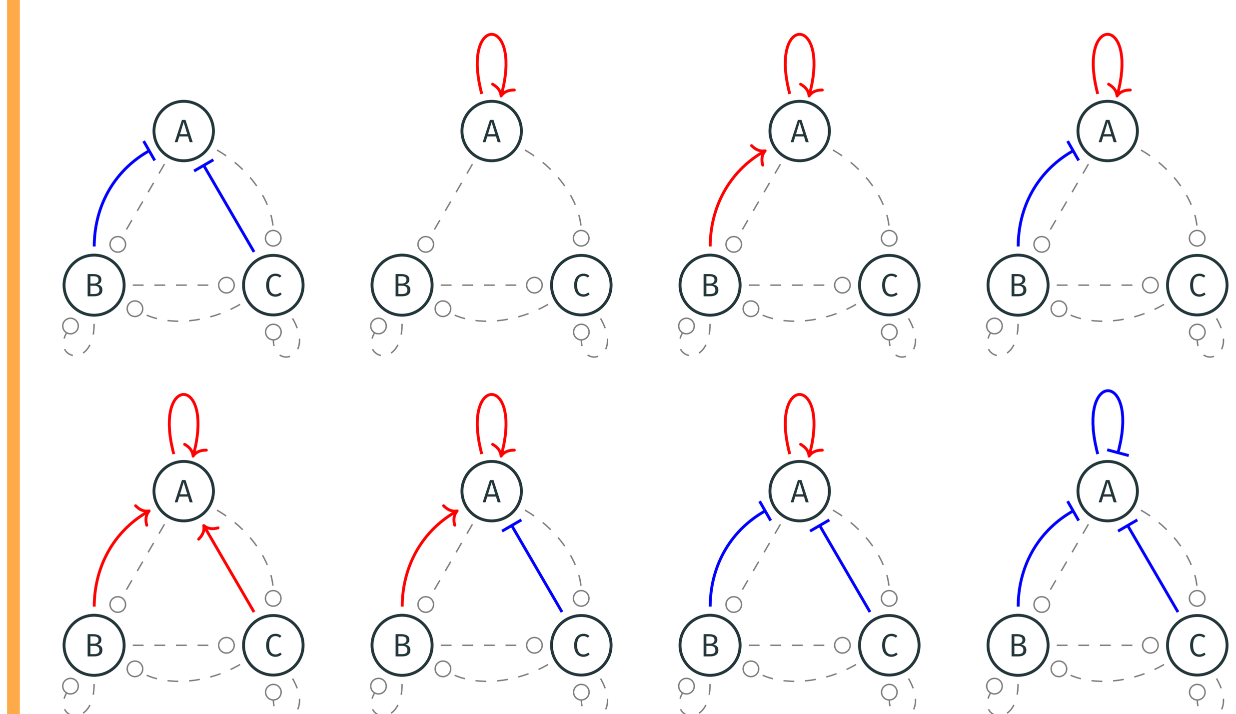


5 node: 1 inhibition in each node -1,979,631,360 *MBMs*



Networks offering Single-High Multistability

No self-edges:
Should be fully connected. All inputs should be inhibitory.
With self-edges:
Self-activation: Any input can be present (sources/sign).
Self-inhibition: Fully connected. All inputs inhibitory.



Multistability iff each vertex v_i has positive self-edge, or all other vertices have repressive edges terminating at v_i.

Conclusion

- Maximize single-high → Team network: A vs rest
- Maximize all single-high → Disconnected
- Equal single-high → Symmetric networks + Degeneracy
- Multistable single-high → Fully connected mutually inhibitory (or) self-activation

Future Directions

- Networks with the most models supporting multistability between single high states.
- Extending formalism to ODE parameter space.

References

- Zhou, J. X., & Huang, S. (2011). Trends Genet.
- BV, H. et al. (2025). J R Soc Interface.
- Gedeon, T. (2024). NPJ Syst Biol Appl.
- Adigwe, S. et al. (2025). bioRxiv.